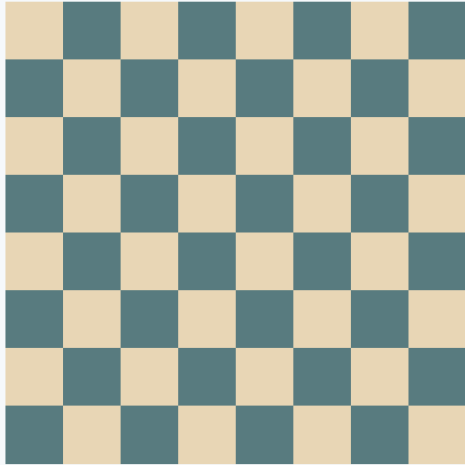




PCOD HYBRID 0.2

A documented, hybrid chess database format

HOT INDEX

Fast metadata & search

NAME DICTIONARY

Stable IDs, less duplication

PAYLOAD STREAM

Full game content & annotations

PCOD HYBRID 0.2

A DOCUMENTED, HYBRID AND EXTENSIBLE CHESS GAME DATABASE FORMAT

An overview for users and chess software developers

Document version 1.0 - June 2026

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Scope

This document describes PCOD Hybrid 0.2 at the architecture, application and integration levels. It is not a compatibility promise for other format versions.

CONTENTS AT A GLANCE

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PCOD Hybrid 0.2 combines a compact database core with PGN interoperability.

1. EXECUTIVE SUMMARY

PCOD Hybrid 0.2 is a folder-based chess game database format designed to combine three goals that are rarely achieved together: fast access to large collections, faithful preservation of PGN content, and an architecture that can be extended comfortably in the Java ecosystem. Instead of placing everything in a text file or a single opaque binary block, PCOD separates hot metadata, recurring names and variable game payloads.

The format uses fixed 56-byte index records for predictable random access; stable-ID, front-coded name dictionaries to reduce duplication; and an append-only payload stream that stores either compact moves or compressed PGN according to game complexity. Optional H2 and sound sidecars can add functions without changing the core index.

PCOD is not intended to replace PGN as the universal exchange language. PGN remains the common denominator between programs. PCOD is intended as a working database: import PGN, browse and search quickly, annotate, teach, analyse, and export back to standard formats.

Its main strengths are a documented structure, PGN fidelity, recovery-oriented updates and extension potential. Its present weaknesses are a young ecosystem, limited third-party support and the need for strict version discipline while the format is still at 0.2.

Key idea

Fast metadata stays in a fixed index; complete chess meaning stays in a recoverable payload.

2. WHY THE FORMAT WAS CREATED

Chess database software has evolved along several paths. Commercial products commonly provide excellent indexing, rich annotation and polished workflows, but their native formats are usually proprietary or tightly connected to the vendor ecosystem. Users can be highly productive inside the product yet face barriers when they want to build their own tools, integrate teaching workflows or move data into a custom system.

Open solutions solve different parts of the problem. PGN is exceptionally portable, while Scid offers a mature open-source application. However, PGN is sequential text and does not include a standard high-performance index. Binary open-source databases are often deeply connected to long-lived project architectures; significant customization may require knowledge of C/C++, Tcl, internal file structures and the full application flow.

PCOD emerged from the need for a middle path: compact and searchable enough for large working collections, while preserving PGN and making the data model understandable to developers using modern Java tooling.

The Open Chess Game Database Standard project - OCGDB - contributed the idea that chess data should be structured, queryable and documented, with names aligned to familiar PGN concepts [1]. PCOD is not binary-compatible with OCGDB and does not copy its SQLite model. It adopts the broader design motivation and implements a separate hybrid binary architecture.

Java was selected for cross-platform deployment, strong standard libraries, binary I/O, compression, Unicode and modular development. The same data model can therefore support a desktop application, command-line tools, batch converters, services and an extension host.

3. DESIGN PRINCIPLES

PCOD Hybrid 0.2 is organized around seven principles:

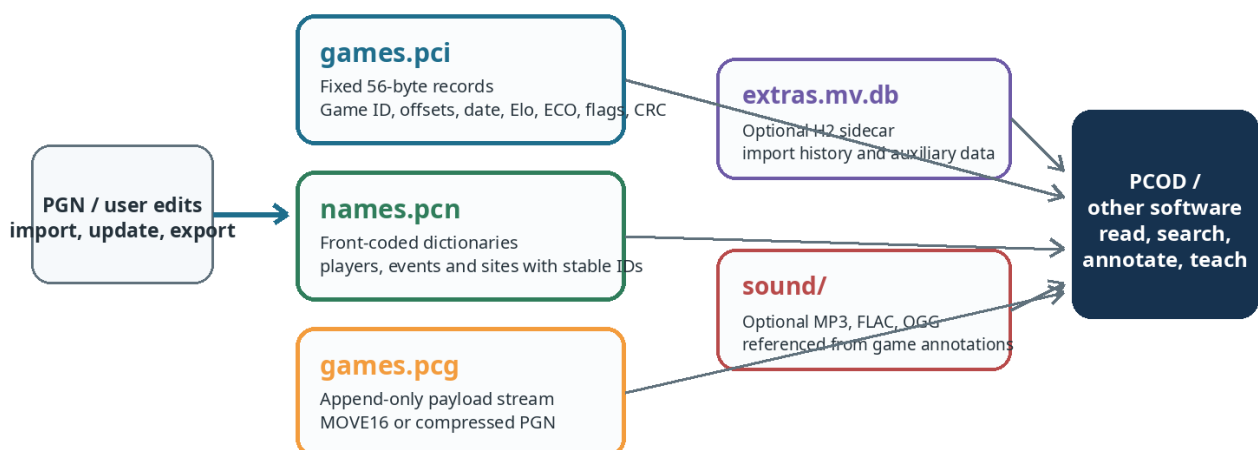
- ◆ **Fast common-path reads:** Game lists and common filters use fixed index records without inflating every PGN payload.
- ◆ **Content fidelity:** Simple linear games can use MOVE16; games with comments, variations, NAGs, special FEN or complex content use compressed PGN.
- ◆ **Stable identifiers:** Game IDs are tied to record positions; player, event and site names retain stable logical IDs.
- ◆ **Append-only payloads:** Updates append new data and change the index pointer only after the new payload is written and checked.
- ◆ **Recovery support:** Import and update markers define recoverable transactions; the committed game count is written last.
- ◆ **Small core, extensible edges:** H2, sound and extension data can be added without changing the 56-byte core index record.
- ◆ **PGN as the bridge:** PGN import and export remain the interoperability path to the wider chess software world.

4. DATA STRUCTURE AND OPERATION

A PCOD database is a directory bundle ending in .pcod, not a single file. The minimum structure contains manifest.properties, games.pci, names.pcn and games.pcg. extras.mv.db, sound/ and recovery/ are optional or temporary components.

manifest.properties identifies the format, version, database UUID, record size and supported codecs. A reader must check formatVersion before applying the 0.2 byte layout.

PCOD Hybrid 0.2 - three-layer architecture



Core data remains compact; optional sidecars add functions without changing the fixed index.

Figure 1. Three-layer architecture and optional sidecars.

games.pci is the hot index: a 64-byte header followed by one 56-byte little-endian record per game. A record contains the payload pointer, name IDs, date, Elo, ECO, result, flags, ply count, semantic hash and payload CRC. A game list can therefore be rendered and filtered without decoding every full game.

names.pcn stores Player, Event and Site dictionaries. Strings use front coding to reduce repetition, while logical IDs remain stable even if physical entries are sorted by name. This keeps the index compact when the same names appear thousands of times.

games.pcg is an append-only payload stream. MOVE16_V1 favours compact storage for sufficiently simple linear games; PGN_DEFLATE_V1 stores compressed UTF-8 PGN for information-rich games. Readers verify CRC32 before decoding.

extras.mv.db is an optional H2 sidecar for import history or auxiliary metadata. sound/ stores MP3, FLAC or OGG files referenced by game annotations. Neither component is required to decode the core chess game.

Component	Primary role	Design characteristic	Practical value
manifest.properties	Database identity	Java Properties with formatVersion	Prevents a reader from applying the wrong layout
games.pci	Hot index	64-byte header, 56-byte records, little-endian	Fast browsing and filtering
names.pcn	Name dictionary	Stable IDs and front coding	Reduces repeated names and index size
games.pcg	Game content	Append-only, MOVE16 or compressed PGN	Compact simple games and faithful complex games
extras.mv.db	Optional sidecar	H2 under MPL 2.0	Auxiliary metadata without changing the core
sound/	Multimedia	Content files referenced by annotations	Audio lessons and spoken commentary
recovery/	Transaction recovery	Import and update markers	Lower risk after interrupted writes

Reading a game

How a game is read and safely updated



Safe update rule

append new payload → write recovery marker → update index pointer → verify → remove marker
 Old payload may remain orphaned until compaction; Game ID and 56-byte record position stay stable.

Figure 2. Read flow and safe update path.

1. Use the Game ID to calculate the record offset in games.pci.
2. Read the 56-byte record and validate offset and length.
3. Resolve whiteId, blackId, eventId and siteId in names.pcn.
4. Read the payload from games.pcg and verify CRC32.
5. Decode MOVE16 or inflate PGN.
6. Build the game model for display, search or export.

Safe update flow

When a game is edited, PCOD does not overwrite the old payload in place. It appends a new payload to games.pcg, creates a recovery marker, validates the new data and then updates the index pointer. The old payload may become orphaned and can be removed during a later compaction pass.

This approach prioritizes recovery and stable Game IDs. The trade-off is that databases with many edits should be compacted periodically.

5. PRACTICAL APPLICATIONS

Personal and club collections

Import many PGNs, filter by player, event, ECO, date, Elo and length, and annotate while retaining PGN export.

Opening and opponent research

Combine a fast index with full payloads to build opening trees, statistics and model-game lists.

Teaching and publishing

Store variations, NAGs, comments, diagram markers, graphics and audio; produce worksheets and teacher answer keys.

Conversion workflows

Use PCOD as a working hub between PGN and HTML, LaTeX, PDF, ODT, Excel or advanced SVG.

Specialized search

Add separate indexes for positional patterns, tactics, pawn structures and educational labels without changing the core index.

Data science and AI

A read-only reader can provide normalized games for statistics, machine learning and exercise generation.

Cross-platform products

A shared Java model can support desktop applications, CLI utilities, batch services and an Extension Host.

6. ADVANTAGES AND LIMITATIONS

Advantages	Limitations
✓ Fixed index records support predictable random access and paging.	• The format is young and lacks the broad ecosystem of PGN, Scid or commercial databases.
✓ Hybrid codecs balance storage size and PGN fidelity.	• A multi-file bundle supports recovery but is less convenient than a single file; sharing normally requires ZIP packaging.
✓ The name dictionary reduces duplication while preserving stable IDs.	• Mixed endianness and two codecs make a correct writer more complex than a reader.
✓ Append-only writes and recovery markers are safer than in-place payload replacement.	• Append-only updates create orphaned payloads and require compaction to reclaim space.
✓ The format is documented, making third-party read-only tools practical.	• Third-party tools must implement manifest checks, CRC, dictionaries and recovery correctly.
✓ Unicode and Java integration support cross-platform development.	• Version 0.2 needs an explicit migration policy for future revisions.
✓ Sidecars and extensions create room for new functions without changing the core record.	• Deep position search depends on additional indexes or extensions, not only the core hot index.
✓ Educational annotations and audio can travel with the database bundle.	• Independent validators, SDKs and mature tooling are still limited.

7. COMPARISON WITH COMMON FORMATS

The following comparison is about data formats and ecosystems rather than a full product ranking. Products and capabilities can change by version. PGN is a public text exchange format [2]; Scid is a GPL multi-platform project [3]; Scid vs. PC documents a three-file .si4/.sn4/.sg4 database with historical limits [4]; ChessBase supports 2CBH/CBH and related formats [5][6]; Chess Assistant uses CDP and supports popular formats [7][8].

Criterion	PCOD Hybrid 0.2	PGN	Scid 5 / SCID5	Scid vs. PC / SI4	ChessBase 2CBH/CBH	Chess Assistant CDP
Model	Hybrid bundle: index + dictionary + payload	Sequential text	Scid project binary database	Three index/name/game files	Commercial binary family	Commercial binary database
Specification openness	Documented 0.2 byte layout	Public and widely implemented	GPL source; format tied to project implementation	Public file-format documentation	Proprietary	Proprietary
Human readability	Low	Very high	Low	Low	Low	Low
Interchange	Currently limited; PGN is the bridge	Very high	Moderate inside Scid ecosystem	Moderate inside SI4 ecosystem	Strong inside vendor ecosystem; external write support varies	Moderate through compatible products
Large-database search	High for metadata; extensible indexes	Low without an external index	High	High	Very high in the ecosystem	Very high in the ecosystem
Rich annotations	Full PGN, graphics and audio directives	Comments, NAGs, variations; extensions vary	Rich in the application	Rich in the application	Very rich multimedia and database text	Rich, product-oriented
Customization	High through Java APIs, sidecars and extensions	High as text; low as a database	High by modifying source, but project knowledge is required	High by modifying legacy source	Low for the native format	Low for the native format
Cross-platform	High with Java	Very high	High	Primarily desktop project packages	Primarily the vendor Windows/web ecosystem	Primarily Windows
Maturity	Emerging	Very mature	Mature	Mature legacy format	Very mature commercial ecosystem	Mature commercial ecosystem
Main limitation	Limited third-party tooling; migration and SDK work remain	Slow large searches and no standard index	Not a neutral exchange standard	Legacy limits and older technology	Vendor lock-in and closed specification	Vendor lock-in and closed specification

Note: “High” performance claims refer to normal use inside each format’s main ecosystem; actual results depend on database size, indexes and software implementation.

PGN

Advantages: universal text interchange, human-readable, supported by nearly every chess program.

Limitations: sequential scanning or an external index is needed for large searches; extended directives differ between applications.

Scid 5 / SCID5

Advantages: open source, cross-platform, strong search and reporting, inspectable code. Limitations: the database model and APIs are primarily understood through the Scid codebase; deep customization requires C/C++/Tcl and project-specific knowledge.

Scid vs. PC / SI4

Advantages: good speed, mature tools and documented three-file structure. Limitations: SI4 retains historical limits, including documented maximum game counts and individual game size [4].

ChessBase 2CBH/CBH

Advantages: large commercial ecosystem, strong indexes and multimedia; 2CBH reduces file count and removes several old limits [6]. Limitations: the core specification is closed and third-party write access depends on products and licences.

Chess Assistant CDP

Advantages: CDP is positioned for fast access to millions of games and the products support several common formats [7][8]. Limitations: the native format is proprietary and external developer tooling is limited.

PCOD Hybrid 0.2

Advantages: documented layout, balance of indexed metadata and PGN fidelity, Java/extension orientation. Limitations: limited adoption, and more validators, SDKs, migrations and independent applications are needed.

8. EXTENSION AND CUSTOMIZATION POTENTIAL

A strategic distinction in PCOD is the separation between the data format and extension code.

Extensions must not write directly to games.pci, names.pcn or games.pcg; they operate through game-model APIs, sidecars or separate directories. This allows the user interface and business functions to evolve without making the database unreadable.

Priority extension groups include resource packs for boards, pieces, chess fonts, text fonts and languages; new annotation types; importers and exporters; search providers; education tools; and reports. Data-only resource packs can work with both JVM and Native Image distributions.

Extension and application potential

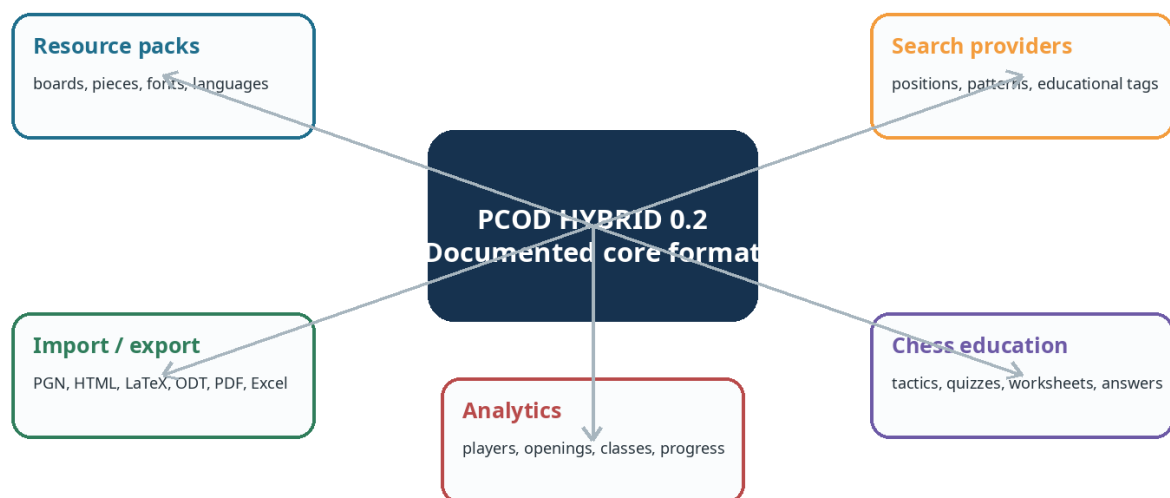


Figure 3. Potential extension ecosystem.

For code extensions, a separate Extension Host communicating through JSON-RPC or a local socket is a safer architecture. It allows extensions to be installed after a Native Image has been built and isolates failures from the main PCOD process.

Other developers can build read-only readers, converters, indexers, integrity checkers, web browsers or education systems. A writer requires full compliance with append-only transactions, CRC validation, stable IDs and recovery rules.

9. GUIDANCE FOR DEVELOPERS

Start with a read-only reader - Parse the manifest, index, dictionary and payload; verify CRC; export PGN for comparison.

Create a fixture suite - Include a simple MOVE16 game, compressed PGN, special FEN, variations, NAGs, audio, soft deletes and recovery cases.

Test round trips - PCOD → PGN → independent parser; compare tags, moves, variations and comments.

Never copy index records between databases - Name IDs are meaningful only inside their own dictionary; decode and re-import instead.

Separate format and implementation licences - The format documentation supports interoperability; source code and SDK use remains subject to their respective licences.

Plan for migration - A reader should reject or route clearly when manifest formatVersion is not 0.2.

Use sidecars for new domains - Education indexes, embeddings, analytics and learner records should not be placed in the core 56-byte record.

Back up the whole bundle - Do not copy only games.pcg or the index; keep sound and sidecar data when the user needs them.

Important

A read-only implementation is much easier to validate. Do not implement writing until crash recovery and round-trip tests are complete.

10. POTENTIAL ROADMAP

Horizon	Focus
Near term	Publish an independent validator, sample databases, Resource and Language Packs, and a Java read-only SDK.
Medium term	Extension Host, official import/export providers, position-search indexes, compaction tools and migration from 0.2 to later versions.
Long term	Multiple independent implementations, web services, distributed synchronization, education schemas, an extension repository, open benchmarks and specification governance.

11. CONCLUSION

PCOD Hybrid 0.2 does not compete with PGN as a universal text exchange format, and it does not yet have the history or ecosystem of Scid, ChessBase or Chess Assistant. Its value is in the combination: a fixed index for speed, hybrid payloads for fidelity, append-only transactions for recoverability and Java architecture for extension.

For users, the format can become a long-lived working collection for research, annotation and education. For developers, the documented byte layout and invariants provide a clearer starting point than reverse-engineering a proprietary format.

Its long-term potential will depend on validators, SDKs, migrations, documentation and independent applications that can read the data. A format becomes a platform only when users can continue to access their information outside the original application.

Final perspective

The goal is not another isolated file format; it is a documented working database that keeps PGN as an exit path.

12. GLOSSARY AND REFERENCES

Term	Meaning
Bundle	A directory treated as one database unit.
Hot index	A compact index holding fields used frequently for browsing and filtering.
Payload	The full or variable-length content of a game.
Append-only	Writing new data at the end rather than replacing old content in place.

Term	Meaning
Front coding	Compressing sorted strings by storing the common prefix length with the previous string.
Sidecar	An auxiliary database or file associated with the bundle without changing the core structure.
CRC32	A checksum for detecting accidental payload corruption, not a security signature.
Semantic hash	A 64-bit FNV-1a content identifier, not a cryptographic checksum.
Soft delete	Marking a record deleted while retaining it until compaction.
Compaction	Rewriting a database to remove orphaned payloads and obsolete records.

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Accessed and reviewed in June 2026. Product formats and capabilities may change; consult the current vendor documentation before implementing compatibility.